Programming for Problem Solving Using Python
Unit-1

Session 3: REPL vs. script vs. Jupyter Notebook, linting with flake8 / pylint

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Session-2 Recap

Python Installation Guide

Windows

Download installer from python.org
Run the installer and check "Add Python to PATH"
Follow the installation steps

macOS

Use Homebrew:

Install Homebrew: `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/`
Install Python: `brew install python`

Alternatively, download from python.org

Session-1 Recap

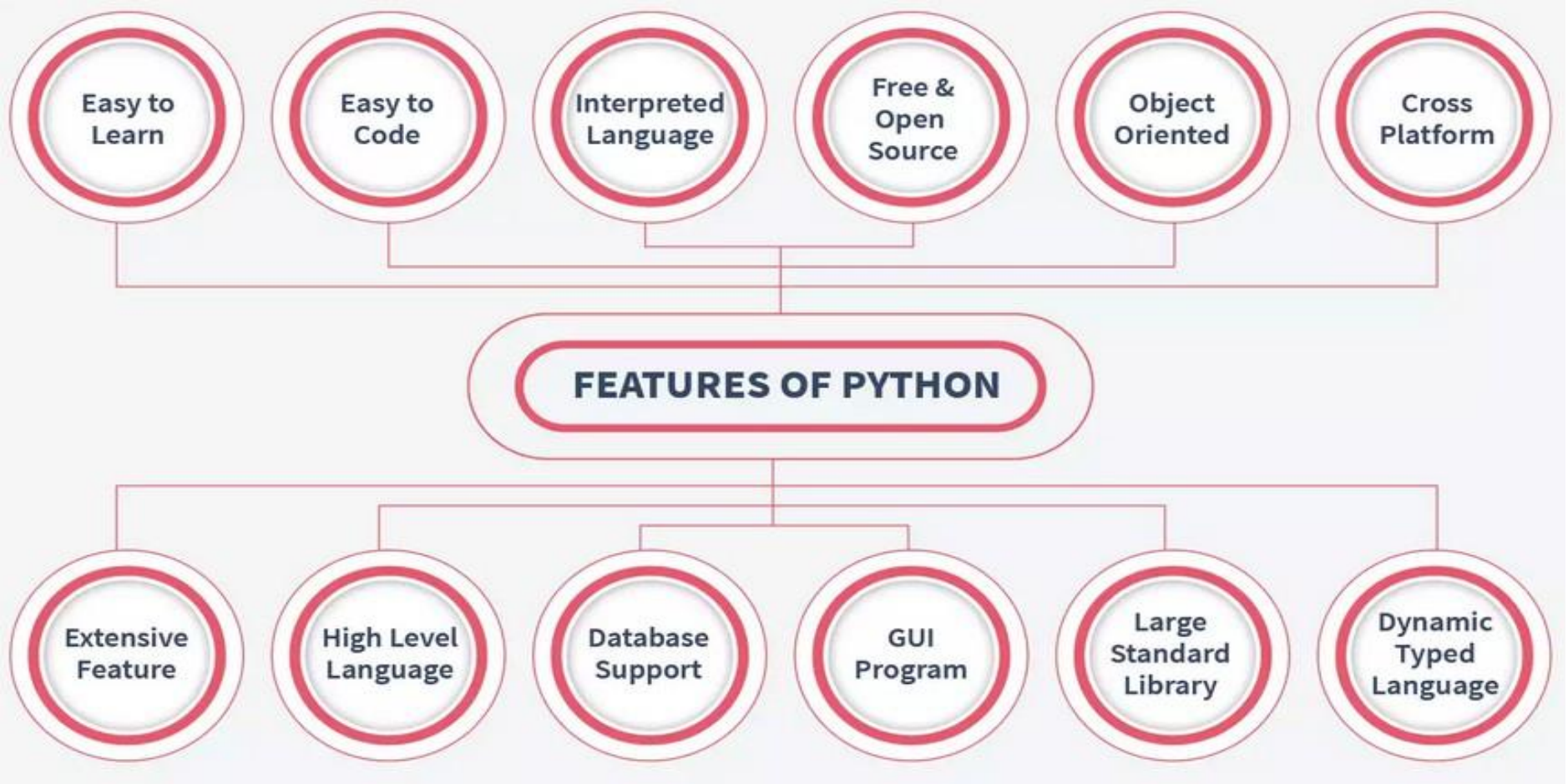
- **Linux**
  - Use package manager:
    - Debian/Ubuntu: `sudo apt-get install python3`
    - Fedora: `sudo dnf install python3`
    - Arch: `sudo pacman -S python`
- **Verify Installation**
  - Open terminal or command prompt
  - Type `python --version` or `python3 --version`

Session-3: Python features, REPL vs. script vs. Jupyter Notebook

Session-3

Python Features

Session-3: Python Features



Session-3: Python Features

1. Easy to Learn

Simple syntax and structure.

Easy for beginners to learn and start coding quickly. Easy to read and understand.

Session-3: Python Features

2. Easy to Code

- Easy Syntax. Comprises words from English language.
- No need to take care of any opening or closing of brackets.
- Uses indentation (spaces or Tabs) for scope.

Session-3: Python Features

2. Easy to Code

- Eg, a simple Python program to add two numbers is as follows:

```
a = 8
b = 9
print(a+b)
```

- Number of lines of code is less in Python.

Session-3: Python Features

3. Interpreted language

- Code is not compiled at once, is converted to a .exe file, and then executed.
- Code is executed line by line and not all at once.
- This line-by-line execution makes it easy to debug the code.

Session-3: Python Features

4. Free and Open Source

- Used for free of cost on any operating system.
- Can be downloaded by anyone from its official web- site.
- Allows making of own modules or libraries.

Session-3: Python Features

5. Object-Oriented Language

- Supports object-oriented programming constructs.
- Easily create and use classes, objects, can implement OOP constructs.

Session-3: Python Features

6. Cross-Platform Language

- Python code written on one machine or operating system can be run anywhere.
- Python code first is converted to an intermediate form known as Bytecode and is then executed.

Session-3: Python Features

7. Extensive feature

- Can be extended to become more versatile.
- Covers a large area in software development applications.
- Is adaptable to various functionalities.
- Can be integrated with other programming language's code.

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Session-3: Python Features

7. Extensive feature

- Makes use of other languages in Python code possible.
- Can easily be embedded in existing code to make it more robust and enhance its features.
- Other languages can be used to compile our python code.

Session-3: Python Features

8. High-Level Language

- Programmers can write codes that are less independent of the specific machine type.
- Has strong abstraction from low-level constructs of system or machine.
- The Developer need not be concerned about architecture or underlying machine type.

Session-3: Python Features

9. Databases Support

- Python Database API (DB-API).
- Databases supported by standard Python:- MySQL, PostgreSQL, Microsoft SQL, Oracle, Informix, etc.
- Interface for the particular database needs to be imported.
- Supports both relational as well as non-relational databases.

Session-3: Python Features

10. GUI Programming

- Supports features and libraries for graphical development.
- System calls and procedures defined to develop an application's perfect GUI.
- Examples of Python frameworks for GUI:- Django, Tkinter, etc.

Session-3: Python Features

11. Large Standard Library

- Bulk of cross-platform libraries, provides rich set of modules and functions.
- Libraries compatible with various operating systems like UNIX, Mac, windows, etc.
- A Large number of libraries support the import and use of required functionality.

Session-3: Python Features

12. Dynamic Typed Language

- No need to declare the data type of a variable explicitly.
- The Data type of variable is decided at run time.
- One variable can be used to store different types of data at different instances in the program.

Session-3: Power of Python Intrepreted Language

Steps of Python Code Execution:-

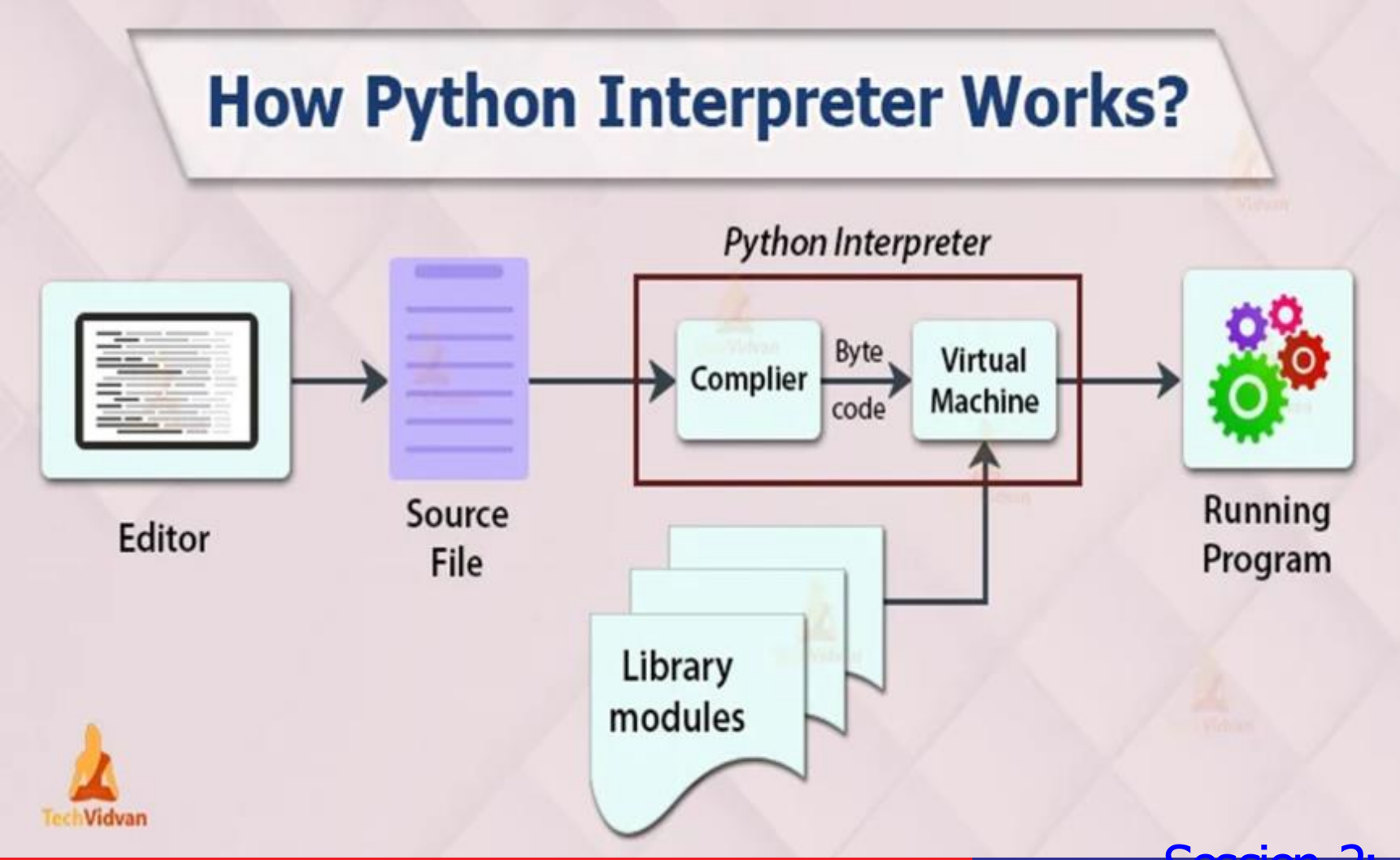
- Converts code into byte code.
- Creates file with extension .pyc
- Internal compilation of byte code, unknown to the developer.

Session-3: Power of Python Intrepreted Language

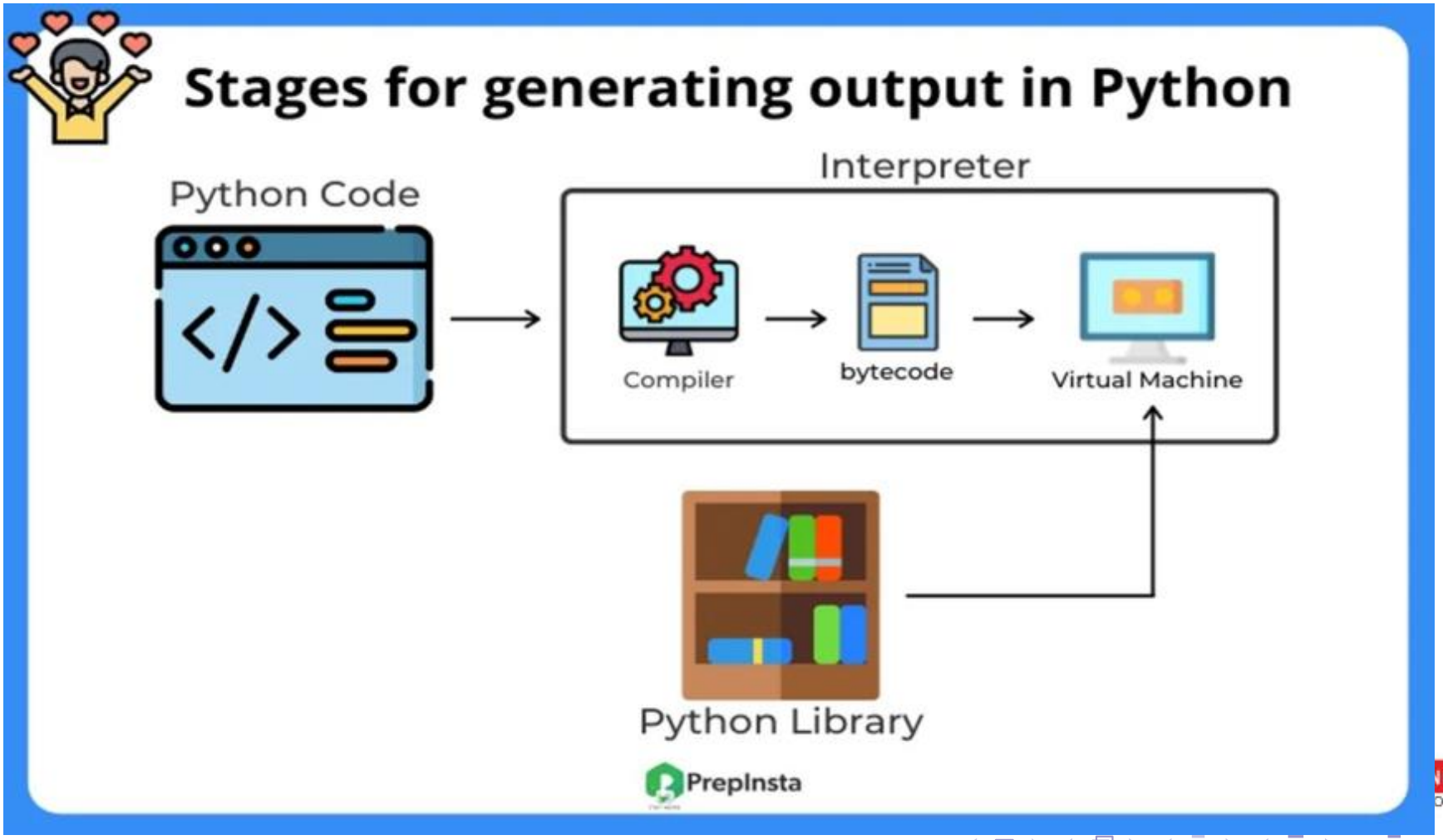
Steps of Python Code Execution:-

- Python Virtual Machine is the interpreter to understand byte code.
- .pyc file, created in the compilation step executed by appropriate virtual machines.
- Virtual Machine is the runtime engine that runs Python scripts.

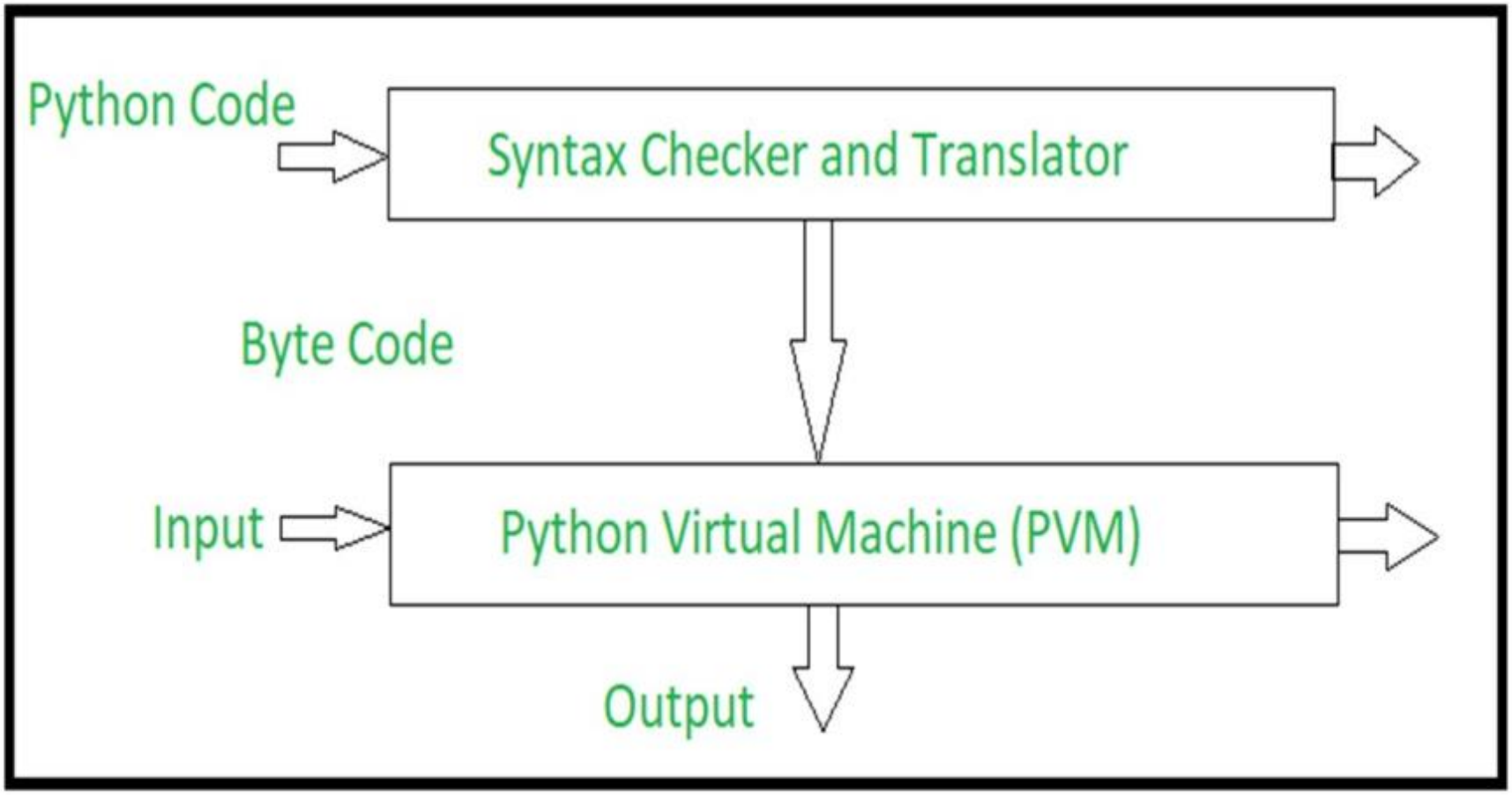
Session-3: Power of Python Intrepreted Language



Session-3: Power of Python Intrepreted Language



Session-3: Power of Python Intrepreted Language



Pythonic Mindset

- **"Pythonic"** means writing code that is **clean**, **readable**, and **efficient**, in line with Python's philosophy.
- Follow the **Zen of Python** (import this) – key principles:
- *Simple is better than complex.*
- *Readability counts.*
- *There should be one — and preferably only one — obvious way to do it.*
- Use built-in functions and idiomatic Python:
- Prefer for item in list: over indexing
- Use list comprehensions instead of manual loops
- Avoid unnecessary complexity – **code should feel natural to read.**

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REPL vs Script vs Jupyter Notebook

Environment	Description	Best For
REPL (Read-Eval-Print Loop)	Interactive command-line mode (python in terminal)	Quick testing & learning
Script (.py)	Python file executed from start to end	Full programs, automation
Jupyter Notebook	Web-based interface with code, text, and visuals	Data science, exploration, teaching

Session-3: Python Applications

Real-world Use Cases/Applications

- *Versatility:* Web development, Data science, Machine learning, Artificial intelligence.
- *Ease of Learning:* Reduced cost of program maintenance.
- *High Demand in the Job Market:* High applicability in various domains.

Session-3: Python Applications

Real-world Use Cases/Applications

- *Data Science and Machine Learning:* Libraries simplify complex tasks.
- *Automation and Scripting:* System administrators, DevOps engineers, and IT professionals use for efficient workflows.
- *Cross-platform Compatibility:* Flexibility

Session-3: Test Your Knowledge- Fill in the blanks

- What is the primary advantage of Python's dynamic typing? _____
- Which feature of Python simplifies the creation of graphical user interfaces (GUIs)? _____
- Python is a what type of programming Language .
- What are some Python frameworks that support GUI? _____.
- Name some of the databases supported by Python _____.

Session-3: Test Your Knowledge- Fill in the blanks (Solutions)

- What is the primary advantage of Python's dynamic typing? Flexibility
- Which feature of Python simplifies the creation of graphical user interfaces (GUIs)?
  GUI frameworks and libraries
- Python is a what type of programming Language
  High-level, Interpreted, General-purpose.
- What are some Python frameworks that support GUI? Tkinter, PyQt.
- Name some of the databases supported by Python
  SQLite, MySQL, PostgreSQL, MongoDB.

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Session-3: Test Your Knowledge- Quiz

- Q1. What is the type of programming language supported by Python?
  1. Object-oriented
  2. Functional programming
  3. Structured programming
  4. All of the above
- Q2. In which language is Python written?
  1. English
  2. PHP
  3. C
  4. All of the above
- Q3. Which one of the following is the correct extension of the Python file?
  1. .py
  2. .python
  3. .p
  4. None of these

Session-3: Test Your Knowledge- Quiz (Solutions)

- Q1. What is the type of programming language supported by Python?
  Ans:- All of the above
- Q2. In which language is Python written?
  Ans:- C
- Q3. Which one of the following is the correct extension of the Python file?
  Ans:- .py

Session-3: Test Your Knowledge-Interview Questions

1. What kinds of applications can Python be used for?
2. List 3 Main advantages of Python?
3. Which programming language is a good choice between Java and Python based on different criteria?
4. Differences between Python 2.x and Python 3.x
5. How Python is interpreted?

Session-3: Test Your Knowledge-Interview Questions (Solutions)

1. Web Development, Data Science and Machine Learning, Automation and Scripting.
2. Readability and Simplicity, Extensive Standard Library and Ecosystem, Cross-Platform Compatibility.
3. Python

Session-3: Test Your Knowledge- Interview Questions (Solutions)

4. In Python 2, print is considered to be a statement and not a function. In Python 3, print is considered to be a function and not a statement.
   In Python 2, integer division can be performed using the '/' operator. In Python 3, integer division can be performed using the '//' operator.
5. Python is an interpreted language, which means that its code is executed by an interpreter rather than being compiled into machine code beforehand.

Thank You!

Any Questions?